

iGPC: Generative Motion Priors for Object-Aware Humanoid Interaction

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Abstract—Humanoid robots operating in unstructured environments must combine robust whole-body control with the ability to perceive and physically interact with surrounding objects. While large-scale human motion data provides powerful priors for natural and versatile humanoid control, effectively transferring such priors to perception-driven object interaction remains challenging. To address this bottleneck, we propose a framework that extends the recently proposed Generative Pretrained Controller (GPC) from general human motion to full-body humanoid–environment interaction. First, we adapt GPC into interaction experts conditioned on scene affordance cues and privileged state information. These experts leverage the pretrained human motion prior while learning task-specific contact behaviors, including reaching toward objects, grasping environmental supports for stabilization, and pushing movable objects. Second, we introduce a perception-driven student that retains the pretrained GPC policy and distills interaction skills from the experts using onboard sensory observations. To bridge the gap between privileged expert observations and sensory inputs, we propose two complementary training objectives that enable effective adaptation of the pretrained motion prior during distillation. Notably, our experiments across multiple whole-body interaction tasks demonstrate that large-scale generative human motion priors provide an effective foundation for learning deployable policies for humanoid interactions in contact-rich real-world environments.

I. INTRODUCTION

Recent advances in humanoid control have demonstrated that large-scale human motion data can provide powerful priors for learning natural and versatile whole-body behaviors. Motion-tracking approaches such as ASAP and Beyond-Mimic transfer agile human motions to humanoid robots [1], [2], while generalist controllers such as GMT and SONIC scale this paradigm to increasingly diverse motion repertoires and command modalities [3], [4]. More recently, generative approaches have gone beyond motion replay by learning reusable distributions over motor behaviors. In particular, GPC [5] learns an autoregressive generative motor prior from large-scale motion data and adapts this prior to downstream control tasks. These results suggest an appealing alternative to learning individual humanoid skills from scratch: pretrain a general motor prior once and adapt it to increasingly complex downstream behaviors.

A key challenge, however, is extending such motion priors from free-space whole-body control to physical interaction with the environment. Reaching for an object, grasping a support to maintain balance, or pushing a heavy object requires the robot to coordinate locomotion, posture, and contact according to its surroundings. Human motion priors provide useful knowledge about coordinated whole-body movement, but do not by themselves tell the robot *where and how to*

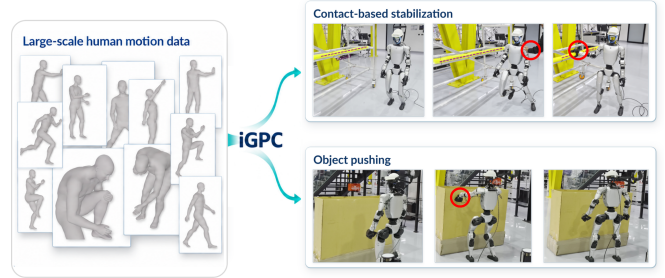


Fig. 1. iGPC adapts human-motion priors to perceptive whole-body interaction on Unitree G1. Real-world sequences illustrate rail-assisted stabilization following an external push (top) and box pushing (bottom).

interact with a particular scene. Recent approaches address perceptive humanoid control through terrain-conditioned motion tracking [6], [7], reference-conditioned object interaction [8], [9], or perception-based loco-manipulation [10], [11], [12]. Nevertheless, it remains unclear how a generative motor prior learned from large-scale human motion can be systematically adapted to object-aware interaction and subsequently retained when the controller must operate from onboard sensing rather than privileged state information.

We address this problem with *iGPC*, a framework that extends *GPC* from generative motion control to object-aware, perception-driven humanoid *interaction*. First, we develop an improved interaction expert that conditions the GPC motion prior on scene affordances describing object geometry, contact, and robot stability. We initialize adaptation using human interaction motions aligned with scene geometry and subsequently optimize task-specific behaviors with reinforcement learning. The resulting experts exploit large-scale human motion priors while learning task-specific interactions, including reaching, grasping environmental supports for stabilization, and pushing movable objects, without requiring reference motions at execution time.

To enable real-world deployment, we propose a perceptive student that retains the same GPC motion prior rather than replacing it with a separately learned action policy. The student uses onboard depth, LiDAR, and proprioception to estimate the interaction and body-state context required by GPC. We introduce distillation objectives that supervise both autoregressive motor-token decisions and resulting joint targets, together with prior-input calibration and task-adaptor refinement. Consequently, the large-scale human motion prior benefits both the experts and perceptive controllers.

Experiments demonstrate strong improvements at both the

expert and student levels. For object-aware reaching, iGPC achieves 99.32% contact success, compared with 31.30% for GPC, while the sensor-based student retains 95.95% success and substantially reduces body collisions. The approach also improves hand-supported recovery from strong disturbances and sustained box pushing, where both the expert and student substantially outperform VisualMimic. Together, these results show that generative human motion priors provide an effective foundation for both privileged interaction learning and perception-driven humanoid control.

In summary, our main contributions are threefold:

- **Object-aware interaction experts.** We extend GPC with scene affordance, contact, and stability conditioning, enabling motion-prior-based reaching, support-assisted stabilization, and object manipulation without reference motions at execution time.
- **Motion-prior-based perceptive control.** We introduce a multimodal distillation framework that transfers privileged experts to onboard depth, LiDAR, and proprioception while retaining the pretrained GPC motor prior and decoder.
- **Extensive experimental validation.** We demonstrate strong performance of iGPC across reaching, stabilization, and box pushing, with ablations validating our expert and student designs and comparisons showing substantial improvements over the baselines.

II. RELATED WORK

Reusable motor representations. Motion priors provide coordinated behavior that can be reused for new objectives. BeyondMimic combines tracking with guided diffusion for goal-directed motion composition, while SONIC connects large-scale tracking to planning and multiple command modalities [2], [4]. BFM-Zero supports task prompting through a behavioral latent representation, and LATENT learns high-level tennis control over a motion-derived latent action space [13], [14]. GPC instead models discrete motor skills autoregressively and introduces conditional low-rank adaptation with supervised and reinforcement learning for physics-based character control [5]. We adopt this foundation-training formulation for G1 rather than proposing a new tokenizer or adaptation rule. Our focus is extending the resulting motor representation to object-aware task control using sensor inputs.

Object-aware and perceptive whole-body control. Perceptive BFM and PGMT address terrain-compatible motion tracking by incorporating local geometry [6], [7]. SceneBot reconstructs scenes around retargeted motions and supplies per-link contact commands [9]; ResMimic learns action residuals around a general tracker to follow coupled humanoid-object trajectories [8]. These formulations address interaction while retaining motion-reference inputs. ViBe also retains a reference-conditioned tracker, learning visual extraction and low-rank decoder adaptations around frozen visual and motor backbones [12]. SplitAdapter targets a different source of variation: object load and robot dynamics, represented by factorized context encoders that modulate

TABLE I
QUALITATIVE COMPARISON OF SELECTED METHODS.

Method	Object manipulation	Reactive hand-supported stabilization	Onboard-perceptive contact	Motion-reference-free execution	Frozen-backbone low-rank adaptation
GPC [5]	–	–	–	✓	✓
BeyondMimic [2]	–	–	–	✓	–
ResMimic [8]	✓	–	–	–	–
ULTRA [10]	✓	–	✓	✓	–
LessMimic [11]	✓	–	✓	✓	–
ViBe [12]	✓	–	✓	o ^v	✓
Bracing for Impact [16]	–	✓ ^s	✓ ^s	✓	N/A
Ours	✓	✓	✓	✓	✓

✓: reported; –: not reported. ^sSimulation only. ^vPartial: ViBe uses a nominal standing reference for dodgeball; locomotion and manipulation remain reference-conditioned.

a frozen manipulation policy [15]. For reactive support, Bracing for Impact uses LiDAR-informed wall bracing for push recovery and is evaluated in simulation [16]. Our task-specific adapters condition a shared generative motor foundation on commands, body-relative surface geometry, contact, and stability, without requiring a prescribed motion reference at execution.

Privileged-to-sensor transfer. ULTRA distills a tracker into a multimodal variational controller using action matching, latent regularization, and reconstruction for tracking and sparse-goal execution [10]. LessMimic combines reference-free interaction with depth-based DAgger distillation, replacing explicit geometric inputs with visual features while retaining the control-head architecture [11]. VIGOR supervises both recovery representations and actions for perceptive fall safety [17]. Retaining motor components while learning their inputs is also established: RMA estimates an environment latent for a retained base policy [18], and AnyBody adapts a sparse-keypoint encoder through latent and action supervision with a frozen decoder [19]. Generalized on-policy distillation supervises learner-generated sequences [20]; we apply this principle to motor-token prefixes generated within a physical control step. Our transfer method reconstructs the physical observation interface and supervises decisions and joint targets through a retained autoregressive controller. A separate bounded calibration map adjusts token-conditioning inputs while leaving the reconstructed physical state supplied to the decoder unchanged. This explicitly separates correction of the controller’s decision inputs from reconstruction of its decoder-state input. Table I compares selected methods by object manipulation, reactive hand support, onboard perception, motion-reference dependence, and motor-backbone adaptation.

III. PROBLEM FORMULATION

Task-conditioned whole-body interaction.

We model task-conditioned whole-body interaction on a Unitree G1 as an episodic partially observable Markov

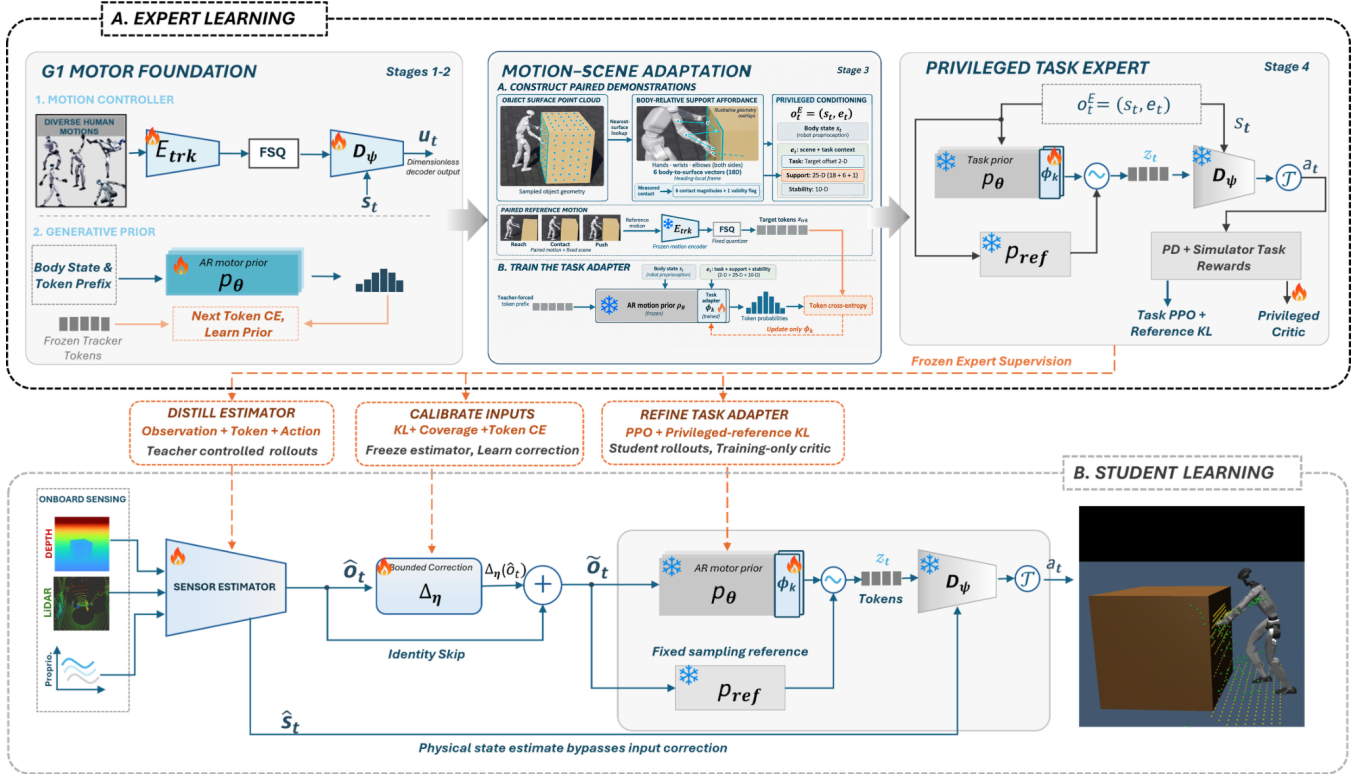


Fig. 2. Overview of the proposed generative control framework. **(A) Expert learning.** A G1-specific motion controller and autoregressive motor prior form the motor foundation. Motion-scene supervision initializes object-conditioned task adapters, which are subsequently refined using privileged simulation observations. **(B) Onboard control.** Depth, LiDAR, and proprioceptive history reconstruct the expert’s observation interface. Observation, token, and action supervision train the estimator, followed by bounded input calibration and task-adapter refinement. Calibrated observations $\tilde{o}_t$ condition token selection, while the physical-state estimate $\hat{s}_t$ bypasses this correction and directly conditions the decoder D_ψ . The original motor-backbone weights and decoder remain fixed during downstream adaptation and sensor transfer. Dashed orange arrows indicate training supervision.

decision process. The hidden state x_t comprises robot configuration and velocities, object states, and contact conditions. Dynamics $x_{t+1} \sim P(\cdot | x_t, a_t)$ include contact interactions and external disturbances. The learned action $a_t \in \mathbb{R}^{29}$ specifies joint-position targets for low-level proportional-derivative control; Dex3 finger control is separate (Sec. V-A). Task conditions are implicit in the policy and reward notation. For each task k , we seek a policy π_k that maximizes

$$J_k(\pi_k) = \mathbb{E}_{\pi_k, P} \left[\sum_{t=0}^{T-1} \gamma^t r_k(x_t, a_t) \right], \quad (1)$$

where T is the episode horizon and $\gamma \in [0, 1)$ is the discount factor. Task-specific rewards encourage task progress, useful contact, and stable execution while penalizing undesirable physical interaction.

Training and deployment observations. During simulation training, o_t^E denotes the complete expert input, comprising the body-state representation s_t , interaction context e_t , and supplied task conditions. The deployed student acts according to

$$a_t \sim \pi_k(\cdot | \mathcal{I}_t), \quad (2)$$

where $\mathcal{I}_t$ includes current depth and LiDAR measurements, finite proprioceptive and previous-action history, onboard state estimates, sensor status, and supplied task conditions.

Recurrent encoders process the finite history window without carrying persistent hidden state between policy calls. Privileged object poses, simulator body velocities, and contact-force labels are not inputs to the learned student body policy.

IV. METHOD

iGPC combines object-aware expert learning (Sec. IV-A) with transfer to onboard sensing (Sec. IV-B), retaining the GPC motor backbone and decoder (Fig. 2). The student reconstructs expert inputs and matches motor-token decisions and joint targets, followed by prior-input calibration and task-adapter refinement.

A. Object-Aware Interaction Experts

Retained motor foundation. We use a G1-specific motor foundation trained following GPC [5], retaining its finite scalar quantization (FSQ) representation, autoregressive prior, and conditional low-rank adaptation mechanism. Let E_{trk} denote the tracking encoder and $z_t = (z_{t,1}, \dots, z_{t,M})$ the motor-token sequence at control step t . The prior $p_{\theta, \phi_k}(\cdot | z_{t, < j}, o_t)$ predicts token j from the preceding tokens and complete controller input o_t , using task-adapter parameters ϕ_k . The decoder converts the tokens and body state into joint-position targets:

$$\begin{aligned} u_t &= D_\psi(s_t, z_t), \\ a_t &= \mathcal{T}(u_t) = q_{\text{off}} + q_{\text{scale}} \odot \tanh(u_t). \end{aligned} \quad (3)$$

Here u_t is the dimensionless decoder output, and q_{off} and q_{scale} are fixed joint offsets and scales. The motion vocabulary, original prior parameters θ , and decoder parameters ψ remain fixed during downstream adaptation.

Interaction and affordance pre-training.

Using the G1-retargeted BONES-SEED [21] collection described in Sec. V-A, we pair each motion with a support scene. Motion metadata and hand/root kinematics guide fixed-support placement around interaction intervals. Walking clips with nearby supports or invalid geometry provide non-interaction examples.

We represent the interaction context as $e_t = (g_t, b_t)$. The support descriptor g_t combines surface geometry, contact magnitudes, and a geometry-validity indicator. For a selected body i , its geometric feature is

$$d_{t,i} = R_t^\top (p_{t,i}^{\text{obj}} - p_{t,i}^{\text{body}}), \quad (4)$$

where the two positions are expressed in world coordinates, and R_t maps pelvis-heading coordinates to the world frame. During Stage-3 rollouts, $p_{t,i}^{\text{obj}}$ is the closest sampled point on a valid support surface. The left/right hands, wrists, and elbows provide six displacement vectors. These are combined with six normalized body-contact force magnitudes and one validity flag to form g_t . Displacements are scaled and clipped before conditioning the controller. The stability descriptor b_t complements these features with torso orientation, ground-relative height, and linear and angular motion.

Training examples are collected through frozen-tracker rollouts in the paired scenes. Each input combines the current simulated body state, support and stability descriptors, and a reference-derived future-root planar target. The tracker encoder and FSQ provide aligned token targets z_t^{trk} , which also drive the decoder during the rollout. Contact features are measured in simulation and may include ground contact; they are not annotated intended-contact labels. This pairing associates motion-derived motor decisions with the geometry and physical state in which they are executed.

Following GPC [5], we optimize only the DoRA/FiLM adapters and conditioning projections with teacher-forced cross-entropy averaged over $M = 8$ tokens and label smoothing 0.01. Geometry, contact, and stability serve as conditioning features, not separate SFT objectives. All motor-foundation components remain fixed.

Privileged task learning. Task reinforcement learning specializes ϕ_k for Eq. 1 using GPC’s token-space optimization. Surface associations, contact definitions, and task conditions are specialized within the same input layout. The retained motor foundation then executes from current state and interaction context without prescribed motion trajectories.

B. Student Distillation and Sensor Transfer

Sensor transfer preserves both token selection from interaction context and action decoding from physical state. We train the reconstructed input interface through the retained expert, then calibrate decision inputs and refine the task adapter. Task-specific training variants appear in Sec. V-A.

1) *Structured Observation Reconstruction:* The structured estimator F_ϑ combines learned perception with fixed kinematic reconstruction:

$$\hat{o}_t = F_\vartheta(\mathcal{I}_t), \quad (5)$$

where $\mathcal{I}_t$ is the available sensor history and supplied task information defined in Sec. III. The reconstructed input contains body state $\hat{s}_t$, interaction context $\hat{e}_t = (\hat{g}_t, \hat{b}_t)$, and supplied task conditions passed through unchanged.

Depth and LiDAR encoders extract local geometry, while a recurrent proprioceptive encoder summarizes motion, effort, tracking error, and previous actions. Their fused features, together with task conditions, sensor status, and ground features, predict body-relative surface displacements, contact features, and geometry validity. Simulator geometry and contact measurements supervise these predictions during training.

We reconstruct the articulated body state using known kinematics rather than predicting every body quantity independently. A separate recurrent estimator predicts torso height and heading-local linear velocity, $\hat{\xi}_t = (\hat{h}_t, \hat{v}_t)$, from proprioceptive history and ground features. Ground features combine LiDAR plane fitting and forward-kinematic foot geometry. The body and stability estimates are then

$$(\hat{s}_t, \hat{b}_t) = \mathcal{K}(q_t, \dot{q}_t, Q_t, \omega_t, \hat{\xi}_t), \quad (6)$$

where $q_t, \dot{q}_t$ are measured joint positions and velocities, Q_t is the pelvis orientation estimate, and ω_t is pelvis-frame angular velocity. The reconstruction map $\mathcal{K}$ accounts for the controller’s height convention, joint-induced motion, and link-origin-to-center-of-mass velocity offsets. Privileged body states provide training supervision but are not inputs to the deployed body policy.

2) *Distillation Through the Retained Prior:* Observation reconstruction alone does not distinguish errors that change motor decisions from those that change physical action decoding. We therefore supervise the estimator at all three levels:

$$\mathcal{L}_{\text{distill}} = \lambda_o \mathcal{L}_{\text{obs}} + \lambda_z \mathcal{L}_{\text{tok}} + \lambda_a \mathcal{L}_{\text{act}}. \quad (7)$$

Here $\mathcal{L}_{\text{obs}}$ is a weighted Smooth- L_1 loss on body state, stability, interaction features, and the base-state estimate $\hat{\xi}_t$. Only estimator parameters ϑ are updated; the task-adapted prior and decoder remain fixed.

Motor-token supervision. Let z_t^E and z_t^S be sequences generated from privileged and estimated inputs at the same simulator state. During distillation, the teacher and student distributions are evaluations of the same fixed task-adapted prior:

$$\begin{aligned} p_E(\cdot \mid z_{t,<j}) &:= p_{\theta, \phi_k}(\cdot \mid z_{t,<j}, o_t^E), \\ p_S(\cdot \mid z_{t,<j}) &:= p_{\theta, \phi_k}(\cdot \mid z_{t,<j}, \hat{o}_t). \end{aligned} \quad (8)$$

For prefix source $r \in \{E, S\}$, we align their next-token distributions after the same prefix:

$$\mathcal{K}_r = \mathbb{E} \left[\frac{\tau^2}{M} \sum_{j=1}^M D_{\text{KL}}(p_E^r(\cdot \mid z_{t,<j}^r) \parallel p_S^r(\cdot \mid z_{t,<j}^r)) \right]. \quad (9)$$

Superscript τ denotes temperature-scaled distributions obtained by dividing logits by $\tau > 0$ before softmax [22]. The token loss also includes expert-sequence cross-entropy:

$$\mathcal{L}_{\text{CE}} = -\mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M \log p_S(z_{t,j}^E | z_{t,<j}^E) \right], \quad (10)$$

$$\mathcal{L}_{\text{tok}} = w_E \mathcal{K}_E + w_S \mathcal{K}_S + w_{\text{CE}} \mathcal{L}_{\text{CE}}.$$

Teacher prefixes supervise the expert’s usual decisions; student prefixes supervise continuation after choices made from estimated context. The latter follows the learner-sequence supervision principle of on-policy distillation [20], applied to motor tokens within one control step.

Decoded-action supervision. Even matching tokens can produce different actions when the decoder receives an inaccurate body state. Define $\hat{u}_t^r = D_\psi(\hat{s}_t, z_t^r)$ and $u_t^E = D_\psi(s_t, z_t^E)$. We supervise decoder outputs and physical joint targets:

$$\mathcal{L}_{\text{act}} = \mathbb{E} [\alpha_E H(\hat{u}_t^E - u_t^E) + \alpha_S H(\hat{u}_t^S - u_t^E) + \alpha_q H_W(\mathcal{T}(\hat{u}_t^S) - \mathcal{T}(u_t^E))]. \quad (11)$$

Here H is a Huber loss and H_W includes joint and sample weights. The teacher-token term isolates decoding error under $\hat{s}_t$; the student-token terms supervise the actions produced by the student’s own decisions.

Teacher targets and sampled tokens are detached. Token losses propagate through the fixed prior into its estimated inputs, while action losses propagate through the fixed decoder into $\hat{s}_t$. No action gradient is propagated through categorical token sampling.

Initial distillation uses teacher-controlled rollouts, with both token-prefix types generated at the same physical states. Calibration and refinement introduce student-controlled episodes and query privileged supervision at visited states in a DAgger-style process [23].

3) *Prior-Input Calibration:* To correct residual token-conditioning errors, we freeze the estimator and apply a bounded residual only to the prior’s inputs, preserving the decoder’s physical-state estimate:

$$\Delta_\eta(\hat{o}_t) = \epsilon_o \odot \tanh(C_\eta(\text{sg}[\hat{o}_t])), \quad (12)$$

$$\tilde{o}_t = \hat{o}_t + \Delta_\eta(\hat{o}_t).$$

The residual network C_η is initialized to output zero, sg denotes stop-gradient, and ϵ_o sets componentwise bounds. Supplied task-conditioning fields have zero correction bounds. Token selection uses $\tilde{o}_t$, whereas the decoder continues to receive $\hat{s}_t$.

GPC’s reference-guided nucleus sampling [5] restricts the active prior to tokens allowed by a frozen reference. Because conditioning errors can alter this set, calibration aligns the reference under privileged and corrected inputs.

For prefix source $r \in \{E, S\}$, let $p_{\text{ref}}^{E,r}$ and $p_{\text{ref}}^{C,r}$ denote the reference distributions conditioned on o_t^E and $\tilde{o}_t$, respectively, after prefix $z_{t,<j}^r$. The reference includes a

fixed adapted-controller snapshot and its observation normalization. Let $\mathcal{N}_{t,j}^{E,r}$ be the top- p nucleus of the privileged reference [24]. We encourage the corrected reference to retain probability mass on this set:

$$\mathcal{L}_{\text{cov}}^r = -\mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M \log \sum_{v \in \mathcal{N}_{t,j}^{E,r}} p_{\text{ref}}^{C,r}(v) \right]. \quad (13)$$

Define $\mathcal{K}_{\text{ref}}^r$ as in Eq. 9, replacing the teacher and student distributions with the privileged and corrected reference distributions. Let $\mathcal{L}_{\text{NLL}}^r$ be the mean negative log-likelihood of the expert token $z_{t,j}^E$ under the active prior conditioned on $(\tilde{o}_t, z_{t,<j}^r)$. The calibration objective is

$$\mathcal{L}_{\text{cal}} = \sum_{r \in \{E, S\}} (\mu_r \mathcal{K}_{\text{ref}}^r + \nu_r \mathcal{L}_{\text{cov}}^r + \zeta_r \mathcal{L}_{\text{NLL}}^r), \quad (14)$$

with nonnegative weights. Only η is updated; the estimator, active prior, reference, and decoder remain fixed.

For execution, write $p_{t,j}^S(v) := p_{\theta, \phi_k}(v | z_{t,<j}^S, \tilde{o}_t)$. The executed distribution $\bar{p}_{t,j}^S$ restricts these probabilities to the reference nucleus computed from $\tilde{o}_t$ and renormalizes them. When their probability mass on that set is below 10^{-3} , our implementation uses reference probabilities renormalized over the same set. Privileged reference evaluations provide training targets only; deployed sampling uses corrected student inputs.

4) *Task-Adapter Refinement:* To optimize task return after calibration, we refine ϕ_k on student-controlled rollouts using GPC’s token-space PPO [5], [25], regularized toward the frozen privileged reference:

$$\mathcal{L}_{\text{RL}} = \mathcal{L}_{\text{PPO}}^S + \beta \mathbb{E} \left[\frac{1}{M} \sum_{j=1}^M D_{\text{KL}}(p_{\text{ref}}^{E,S} \| p_{t,j}^S) \right]. \quad (15)$$

Here $\mathcal{L}_{\text{PPO}}^S$ is the negative clipped PPO surrogate on student-controlled transitions. The reference uses privileged input o_t^E , while the active prior uses corrected input $\tilde{o}_t$; both are evaluated after the same student prefix. PPO likelihoods use $\bar{p}_{t,j}^S$, including the low-overlap fallback, whereas the KL term compares unrestricted categorical distributions.

Policy and privileged-critic updates use student-controlled transitions; reference supervision may also use teacher transitions. All components except the task adapter and critic remain fixed. Observation and action reconstruction losses are discontinued; the critic and privileged observations are used only during training.

V. EXPERIMENTS

A. Experimental Setup

Implementation details. We use a Unitree G1 humanoid with a 29-DoF learned body controller operating at 50 Hz with joint-level PD control. Simulation is performed in Isaac Lab at 200 Hz with a control decimation of four. For rail-grasping experiments, Dex3 hands provide an additional

14 actuated finger joints controlled separately from the learned whole-body policy. In simulation, finger closure is triggered when the selected palm establishes rail contact above 1 N, excluding contact with the rail underside. The finger controller uses simulator-provided palm and finger contact forces, together with hand-rail relative motion, to regulate capture and holding. Excessive contact force causes incremental opening, and episode reset clears the grasp state.

Sensor-based students use 96×64 depth images, 16×180 LiDAR rasters, 25 frames of proprioceptive history, and onboard state estimates to reconstruct the expert inputs defined in Sec. III.

Pre-Training. iGPC is pre-trained on interaction sequences comprising 1,024 clips, totaling approximately 2.24 hours, selected from the Unitree G1-retargeted motions in BONES-SEED [21]. It contains 512 interaction clips spanning wall support, pushing, vaulting, and ground rising, together with 512 walking clips. Interaction pre-training uses this selected motion collection to adapt the motor prior through frozen-tracker token supervision.

Affordance pre-training pairs each clip with a support scene as described in Sec. IV-A. We use a temporal fall-back when annotations are unavailable. Walking examples include 256 scenes with nearby supports and 256 with invalid padding geometry.

Training. The reaching and box-pushing students are trained using observation, token, and action supervision, followed by prior-input calibration and task-adaptor refinement, see Sec. IV-B. The Dex3 student uses a task-specific continuation of this procedure. All reported student results correspond to autonomous student execution.

Evaluation protocol. Reaching and student ablations use 2,048 matched trials per policy, with four evaluation seeds and 512 trials per seed. Each trial lasts 512 control steps (10.24 s). Seeds vary evaluation conditions, not independently trained policies. Policies share initial robot states, rail configurations, and target-hand commands; rail orientation, height, and robot-rail displacement vary across trials. Training sensor corruption, reset noise, action delay, dynamics randomization, and task-specific training terminations are disabled, and no external pushes are applied. Reactive stabilization and box pushing use the protocols below.

B. Object-Aware Conditioning for Commanded Reaching

We evaluate object-aware conditioning by commanding a specified hand to contact the upper crown of a fixed rail across varying positions, orientations, and heights. We compare iGPC with target-conditioned GPC, GPC adapted to interaction motions without affordance observations, and a frozen MaskedMimic [26] controller supplied with a privileged, geometry-derived hand-target trajectory and no task-specific PPO adaptation.

To compare performance we evaluate *upper-crown contact rate* that measures trials with a selected-hand crown contact of at least 15 N. *Body collision* measures pelvis, torso, or head rail contacts above 5 N. *Endpoint error* is the terminal hand-to-target distance, reported as median/p90 over all

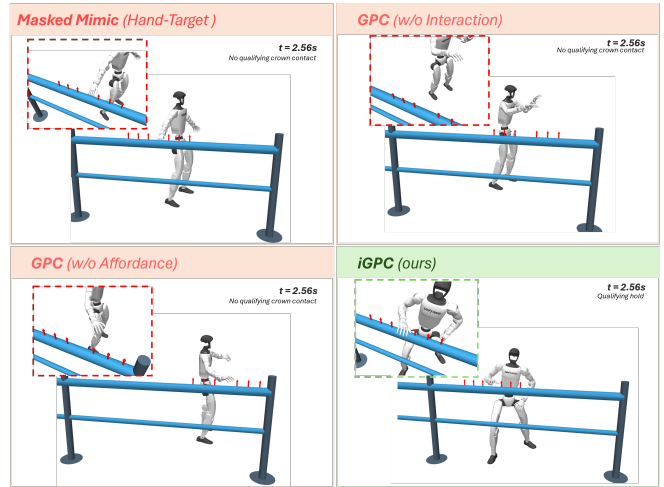


Fig. 3. **Qualitative comparison of commanded rail-contact acquisition.** Snapshots at $t = 2.56$ s compare the policies evaluated in Table II: MaskedMimic hand targeted to rail (top left), GPC (top right), GPC w/o support observation (bottom left), iGPC (ours, bottom right).

TABLE II
EXPERT POLICY EVALUATION FOR COMMANDED RAIL-CONTACT TASK.

Policy	Interaction pre-training	Affordance pre-training	Crown contact (%) $\uparrow$	Endpoint error p50/p90 (cm) $\downarrow$	Time to crown p50/p90 (s) $\downarrow$
MaskedMimic (Hand Target)	—	—	0.20	12.07/15.55	1.45/2.75
GPC (w/o Interaction)	×	×	31.30	16.03/36.43	4.90/9.00
GPC (w/o Affordance)	✓	×	48.63	15.03/32.88	3.93/8.44
iGPC (ours)	✓	✓	99.32	1.35/11.04	0.76/1.46

trials. *Time to crown* is the time to first qualifying contact, reported over successful trials.

Table II shows 99.32% rail-contact success for iGPC, compared with 31.30% without interaction pre-training and 48.63% without affordance conditioning. iGPC reduces median endpoint error from 16.03 to 1.35 cm relative to target-conditioned GPC and reaches the rail faster. MaskedMimic succeeds in 0.20% of trials, so its contact-time statistics summarize only four successes. These comparisons evaluate complete pipelines rather than isolate pre-training effects. Fig. 3 illustrates the resulting hand-rail configurations.

C. Perceptive Student Policy

We evaluate transfer to onboard sensing while retaining the pretrained GPC motor prior. We compare the privileged expert, a direct-action MLP student, and iGPC variants that ablate observation supervision, token supervision, or prior-input calibration. The MLP and no-calibration students use the same sensor history and PPO refinement as full iGPC. The no-calibration variant retains full-loss distillation but omits the learned prior-input residual. All students execute autonomously from sensor observations.

As shown in Table III, the full iGPC student achieves 95.95% contact success, retaining most of the expert’s 99.32% performance while outperforming the expert in terms of body collisions. In contrast, distillation of an MLP student that preserves no motion prior achieves only 46.53% success. When ablating the design choices of our iGPC student, we

TABLE III

PERCEPTION-BASED STUDENT POLICY EVALUATION FOR COMMANDED RAIL-CONTACT TASK.

Policy	Obs. loss	Token loss	Crown contact (%) $\uparrow$	Body collision (%) $\downarrow$	Endpoint error p50/p90 (cm) $\downarrow$	Teacher-target MAE (rad) $\downarrow$
Expert iGPC	—	—	99.32	15.19	1.35/11.04	—
Student MLP	—	—	46.53	50.73	11.19/86.78	0.178396
iGPC (no obs. or token)	$\times$	$\times$	25.39	17.19	51.78/146.20	0.127518
iGPC (no obs.)	$\times$	$\checkmark$	93.41	13.13	1.37/8.38	0.088503
iGPC (no token)	$\checkmark$	$\times$	90.14	10.55	2.68/16.22	0.088679
iGPC (no prior calibration)	$\checkmark$	$\checkmark$	91.02	10.79	1.62/13.53	0.085493
iGPC (full)	$\checkmark$	$\checkmark$	95.95	5.22	1.22/5.62	0.085331

TABLE IV

REACTIVE STABILIZATION UNDER TWO-PUSH DISTURBANCES ON 512 MATCHED TRIALS PER VELOCITY BAND.

Push Δv (m/s)	Policy	Physical fall $\downarrow$	Impact > 500 N $\downarrow$
1.5–2.0	iGPC expert	3 (0.59%)	24 (4.69%)
1.5–2.0	GPC leg-only	12 (2.34%)	34 (6.64%)
1.5–2.0	Student iGPC	5 (0.98%)	55 (10.74%)
2.0–2.5	iGPC expert	24 (4.69%)	91 (17.77%)
2.0–2.5	GPC leg-only	94 (18.36%)	118 (23.05%)
2.0–2.5	Student iGPC	25 (4.88%)	114 (22.27%)
2.5–3.0	iGPC expert	116 (22.66%)	206 (40.23%)
2.5–3.0	GPC leg-only	207 (40.43%)	210 (41.02%)
2.5–3.0	Student iGPC	88 (17.19%)	201 (39.26%)

observe that removing both observation and token supervision reduces success to 25.39%, while removing either component individually yields 90.14–93.41%. Prior calibration further improves success from 91.02% to 95.95% and reduces the 90th-percentile endpoint error from 13.53 cm to 5.62 cm.

D. Whole-Body Interaction Experiments

Reactive stabilization. The humanoid is subjected to two external pushes while a rail is available for hand support. The policy must reactively reach and grasp the rail to maintain balance, without receiving a prescribed reaching motion. We evaluate three disturbance ranges, with push-induced velocity changes of 1.5–2.0, 2.0–2.5, and 2.5–3.0 m/s, using 512 matched trials per range. We compare the privileged Dex3 expert, the sensor-based Dex3 student, and a leg-only GPC baseline. Physical fall rate measures failure to maintain balance, while impact rate reports trials containing body impacts above 500 N.

As shown in Table IV, at 2.0–2.5 m/s, leg-only GPC falls in 18.36% of trials, whereas the expert and student reduce this to 4.69% and 4.88%, respectively. Under the strongest 2.5–3.0 m/s disturbances, the student achieves a 17.19% fall rate compared with 40.43% for leg-only GPC, demonstrating the benefit of exploiting environmental support for whole-body recovery. Fig. 4 illustrates the leg-only GPC policy and the iGPC student policy when subjected to a disturbance near the rail.

Box pushing. The humanoid continuously pushes a 3.8 kg cube over 60 s with effective box–floor friction $\mu = 0.6$. Each policy uses 512 trials across four evaluation seeds (128

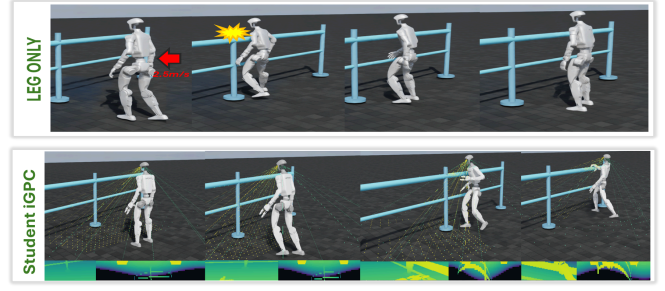


Fig. 4. Reactive stabilization in simulation. Top: leg-only GPC. Bottom: iGPC student with sensor visualizations. The red arrow indicates the applied disturbance.

TABLE V

BOX-PUSHING PERFORMANCE. MEAN $\pm$ SD ACROSS FOUR EVALUATION-SEED MEANS, WITH 128 TRIALS PER SEED.

Policy	Forward displacement (m) $\uparrow$	Absolute lateral drift (m) $\downarrow$
iGPC expert	66.08 $\pm$ 0.15	12.76 $\pm$ 1.35
iGPC student	67.59 $\pm$ 1.05	13.77 $\pm$ 1.08
VisualMimic	45.17 $\pm$ 0.07	36.29 $\pm$ 0.02

trials each). iGPC runs in Isaac Lab and the released VisualMimic [27] controller in MuJoCo. We match shared reset components and box conditions, while each method retains its native robot and control interface. Forward displacement and absolute lateral drift are measured at the first robot fall or 60 s.

As shown in Table V, the iGPC expert and student achieve 66.08 m and 67.59 m mean forward displacement, respectively, compared with 45.17 m for VisualMimic. iGPC policies also substantially reduces lateral drift (12.76 m and 13.77 m versus 36.29 m). These results show that iGPC extends beyond transient contact acquisition to sustained whole-body object interaction.

E. Deployment to the Real Robot

The robot Unitree G1 receives depth images from an Intel RealSense D435i camera at 30 Hz and LiDAR measurements from a Livox MID-360 at 10 Hz. The sensor-reading loop operates at 50 Hz. Proprioceptive inputs include joint positions and velocities, estimated motor torques, IMU-derived orientation and angular velocity, previous joint-position targets, joint-tracking errors, and their history.

Policy inference runs on an NVIDIA RTX 5090 GPU, taking approximately 18 ms per update during hardware execution. The controller operates continuously in closed loop, sustaining a measured wall-clock policy-update rate above 50 Hz. The resulting joint-position targets are sent to the robot's low-level proportional–derivative controllers.

For commanded reaching, the hand-selection condition directs the robot to establish rail contact with the requested hand. Fig. 5 compares example executions of the direct-action MLP and iGPC student. In box pushing, the robot establishes contact, sustains pushing for 10 s, and subsequently releases contact. For reactive stabilization, the robot grasps

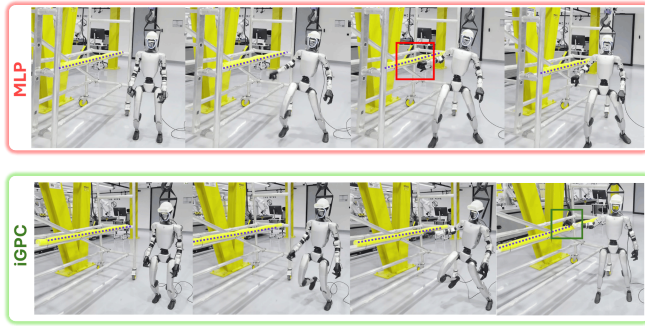


Fig. 5. Real-robot rail-reaching comparison on Unitree G1. Top: direct-action MLP student. Bottom: iGPC student.

the rail in response to an external push, while remaining in stable standing when no disturbance is applied.

VI. CONCLUSION

We presented iGPC for object-aware humanoid interaction and transfer to onboard sensing, retaining the pretrained motor backbone and decoder. Simulation results support iGPC across the evaluated reaching, stabilization, and box-pushing tasks. Broader evaluation of generalization and contact safety remains future work.

REFERENCES

- [1] T. He, J. Gao, W. Xiao, Y. Zhang, Z. Wang, J. Wang, Z. Luo, G. He, N. Sobanbabu, C. Pan, Z. Yi, G. Qu, K. Kitani, J. K. Hodgins, L. Fan, Y. Zhu, C. Liu, and G. Shi, “ASAP: Aligning simulation and real-world physics for learning agile humanoid whole-body skills,” in *Proceedings of Robotics: Science and Systems*, 2025.
- [2] Q. Liao, T. E. Truong, X. Huang, Y. Gao, G. Tevet, K. Sreenath, and C. K. Liu, “BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion,” *arXiv preprint arXiv:2508.08241*, 2025.
- [3] Z. Chen, M. Ji, X. Cheng, X. Peng, X. B. Peng, and X. Wang, “GMT: General motion tracking for humanoid whole-body control,” *arXiv preprint arXiv:2506.14770*, 2025.
- [4] Z. Luo, Y. Yuan, T. Wang, C. Li, F. Castañeda, S. Chen, Z.-A. Cao, J. Li, D. Minor, Q. Ben, J. Park, D. Sami, Z. Wang, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. J. Fan, and Y. Zhu, “SONIC: Supersizing motion tracking for natural humanoid whole-body control,” *Science Robotics*, vol. 11, no. 117, p. eaed4592, 2026. [Online]. Available: <https://arxiv.org/abs/2511.07820>
- [5] Y. Shi, Y. Jiang, C. Tessler, and X. B. Peng, “GPC: Large-scale generative pretraining for transferable motor control,” in *ACM SIGGRAPH 2026 Conference Papers*. Association for Computing Machinery, 2026.
- [6] Z. Wang, Y. Li, T. Ma, Q. Zhang, Y. Fan, H. Xu, S. Yang, and J. Liang, “Perceptive behavior foundation model: Adapting human motion priors to robot-centric terrain,” *arXiv preprint arXiv:2606.08059*, 2026.
- [7] H. Li, P. Li, Y. Tao, Z. Wang, F. Xu, J. Chen, Y. Yuan, D. Jia, Y. Jin, M. Fan, G. Sartoretti, and H. Wang, “PGMT: Perceptive general motion tracking for humanoid robots,” *arXiv preprint arXiv:2609.08511*, 2026.
- [8] S. Zhao, Y. Ze, Y. Wang, C. K. Liu, P. Abbeel, G. Shi, and R. Duan, “ResMimic: From general motion tracking to humanoid whole-body loco-manipulation via residual learning,” *arXiv preprint arXiv:2510.05070*, 2025.
- [9] S. Chen, S. Zhao, Z. Wu, J. Li, G. Shi, and C. K. Liu, “SceneBot: Contact-prompted general humanoid whole body tracking with scene-interaction,” *arXiv preprint arXiv:2606.27581*, 2026.
- [10] X. He, S. Xu, X. Li, R. Dong, L. Bian, Y.-X. Wang, and L.-Y. Gui, “ULTRA: Unified multimodal control for autonomous humanoid whole-body loco-manipulation,” *arXiv preprint arXiv:2603.03279*, 2026.
- [11] Y. Lin, J. Cui, Y. Li, B. Jia, Y. Zhu, and S. Huang, “LessMimic: Long-horizon humanoid interaction with unified distance field representations,” *arXiv preprint arXiv:2602.21723*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.21723>
- [12] L. Krishna, S. Venkatesan, A. Zhang, and Q. Nguyen, “ViBe: Visual behavior adaptation for perceptive humanoid whole-body control,” *arXiv preprint arXiv:2609.09918*, 2026.
- [13] Y. Li, Z. Luo, T. Zhang, C. Dai, A. Kanervisto, A. Tirinzoni, H. Weng, K. Kitani, M. Guzek, A. Touati, A. Lazaric, M. Pirota, and G. Shi, “BFM-Zero: A promptable behavioral foundation model for humanoid control using unsupervised reinforcement learning,” in *International Conference on Learning Representations*, 2026. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2026/hash/80b677bb4d17f77220be88e1d1fdebd-Abstract-Conference.html
- [14] Z. Zhang, H. Lu, Y. Lian, Z. Chen, Y. Liu, C. Lin, H. Xue, Z. Zeng, Z. Qi, S. Zheng, Q. Luan, J. Wang, J. Xing, H. Wang, and L. Yi, “Learning athletic humanoid tennis skills from imperfect human motion data,” *arXiv preprint arXiv:2603.12686*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.12686>
- [15] J. Kang, H. Cho, S. Kang, and D. Koo, “SplitAdapter: Load-aware humanoid loco-manipulation via factorized adaptation,” *arXiv preprint arXiv:2606.03297*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.03297>
- [16] L. Yang, B. Werner, A. B. Ghansah, and A. D. Ames, “Bracing for impact: Robust humanoid push recovery and locomotion with reduced order models,” in *2025 IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids)*, 2025, pp. 728–735.
- [17] O. Azulay, Z. Xu, A. Scheffer, and S. X. Yu, “VIGOR: Visual goal-in-context inference for unified humanoid fall safety,” *arXiv preprint arXiv:2602.16511*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.16511>
- [18] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Proceedings of Robotics: Science and Systems XVII*, Virtual, Jul. 2021.
- [19] S. Li, S. Li, J. Li, and M. Ding, “AnyBody: Free-form whole-body humanoid control from arbitrary keypoint guidance,” *arXiv preprint arXiv:2606.29209*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.29209>
- [20] R. Agarwal, N. Vieillard, Y. Zhou, P. Stanczyk, S. Ramos Garea, M. Geist, and O. Bachem, “On-policy distillation of language models: Learning from self-generated mistakes,” in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/5be69a584901a26c521c2b51e40a4c20-Abstract-Conference.html
- [21] Bones Studio, “Bones-seed: Skeletal everyday embodiment dataset,” <https://bones.studio/datasets/seed>, 2026, available on Hugging Face at bones-studio/seed.
- [22] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015, presented at the NIPS 2014 Deep Learning Workshop. [Online]. Available: <https://arxiv.org/abs/1503.02531>
- [23] S. Ross, G. J. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 15. PMLR, 2011, pp. 627–635.
- [24] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, “The curious case of neural text degeneration,” in *International Conference on Learning Representations*, 2020. [Online]. Available: <https://arxiv.org/abs/1904.09751>
- [25] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017. [Online]. Available: <https://arxiv.org/abs/1707.06347>
- [26] C. Tessler, Y. Guo, O. Nabati, G. Chechik, and X. B. Peng, “Masked-Mimic: Unified physics-based character control through masked motion inpainting,” *ACM Transactions on Graphics*, vol. 43, no. 6, pp. 209:1–209:21, 2024.
- [27] S. Yin, Y. Ze, H.-X. Yu, C. K. Liu, and J. Wu, “VisualMimic: Visual humanoid loco-manipulation via motion tracking and generation,” *arXiv preprint arXiv:2509.20322*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.20322>